Homework 1 1/21/26

Question 1: (Murphy 1.2)

$C_3H_5(NO_3)_3 \longrightarrow CO_2 + H_2O + N_2 + O_2$

$\nu_1 C_3H_5(NO_3)_3 \longrightarrow \nu_2 CO_2 + \nu_3 H_2O + \nu_4 N_2 + \nu_5 O_2$

C: $3\nu_1 \longrightarrow \nu_2 + 0 + 0 + 0$ $\quad \nu_1 = 1 \quad -1\nu_2 + 0 + 0 + 0 = 3$

H: $5\nu_1 \longrightarrow 0 + 2\nu_3 + 0 + 0$ $\quad 0 - 2\nu_3 + 0 + 0 = 5$

N: $3\nu_1 \longrightarrow 0 + 0 + 2\nu_4 + 0$ $\quad 0 + 0 - 2\nu_4 + 0 = 3$

O: $9\nu_1 \longrightarrow 2\nu_2 + \nu_3 + 0 + 2\nu_5$ $\quad -2\nu_2 - \nu_3 + 0 - 2\nu_5 = 9$

$$\underbrace{\begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & -2 & 0 \\ -2 & -1 & 0 & -2 \end{pmatrix}}_{A} \begin{pmatrix} \nu_2 \\ \nu_3 \\ \nu_4 \\ \nu_5 \end{pmatrix} = \underbrace{\begin{pmatrix} 3 \\ 5 \\ 3 \\ 9 \end{pmatrix}}_{B} \qquad \nu = A^{-1} \times B = \begin{pmatrix} -3 \\ -2.5 \\ -1.5 \\ -0.25 \end{pmatrix}$$

Balanced equation: $\underline{1}\, C_3H_5(NO_3)_3 \longrightarrow \underline{3}\, CO_2 + \underline{2.5}\, H_2O + \underline{1.5}\, N_2 + \underline{0.25}\, O_2$

No fractions/decimals: $\boxed{\underline{4}\, C_3H_5(NO_3)_3 \longrightarrow \underline{12} CO_2 + \underline{10} H_2O + \underline{6} N_2 + \underline{1} O_2}$

P1: 15/15

Question 2: (Murphy 1.15)

$\nu_1 HNO_3 + \nu_2 CH_3OH \longrightarrow \nu_3 C_3H_7NO_2 + \nu_4 CO_2 + \nu_5 H_2O$

C: $0 + \nu_2 \longrightarrow 3\nu_3 + \nu_4 + 0$ $\quad \nu_1 = 1 \quad -\nu_2 + 3\nu_3 + \nu_4 + 0 = 0$

H: $\nu_1 + 4\nu_2 \longrightarrow 7\nu_3 + 0 + 2\nu_5$ $\quad -4\nu_2 + 7\nu_3 + 0 + 2\nu_5 = 1$

N: $\nu_1 + 0 \longrightarrow \nu_3 + 0 + 0$ $\quad 0 + \nu_3 + 0 + 0 = 1$

O: $3\nu_1 + \nu_2 \longrightarrow 2\nu_3 + 2\nu_4 + \nu_5$ $\quad -\nu_2 + 2\nu_3 + 2\nu_4 + \nu_5 = 3$

$$\underbrace{\begin{pmatrix} -1 & 3 & 1 & 0 \\ -4 & 7 & 0 & 2 \\ 0 & 1 & 0 & 0 \\ -1 & 2 & 2 & 1 \end{pmatrix}}_{A} \begin{pmatrix} \nu_2 \\ \nu_3 \\ \nu_4 \\ \nu_5 \end{pmatrix} = \underbrace{\begin{pmatrix} 0 \\ 1 \\ 1 \\ 3 \end{pmatrix}}_{B} \qquad \nu = A^{-1} \times B = \begin{pmatrix} 10/3 \\ 1 \\ 1/3 \\ 11/3 \end{pmatrix}$$

Balanced equation (no fractions): $\boxed{\underline{3}\, HNO_3 + \underline{10}\, CH_3OH \longrightarrow \underline{3}\, C_3H_7NO_2 + \underline{1}\, CO_2 + \underline{11}\, H_2O}$

54 mg/L − 10 mg/L = 44 mg/L

$\frac{44\ \text{mg}}{\text{L}} \times \frac{10\ \text{L}}{} \times \frac{1\ \text{g}}{1000\ \text{mg}} \times \frac{1\ \text{mol}\ HNO_3}{63.01\ \text{g}} \times \frac{10\ \text{mol}\ CH_3OH}{3\ \text{mol}\ HNO_3} \times \frac{32.04\ \text{g}}{1\ \text{mol}\ CH_3OH} = \boxed{0.746\ \text{g}\ CH_3OH}$

P2: 20/20

Question 3: (Murphy 1.17) H: 1.008 g/mol ; C: 12.01 g/mol O: 16.00 g/mol

Ethanol (C_2H_6O): $\frac{6(1.008)}{2(12.01)+6(1.008)+16.00} \times 100 = \boxed{13.13\ \text{wt}\%}$

Methane (CH_4): $\frac{4(1.008)}{12.01+4(1.008)} \times 100 = \boxed{25.13\ \text{wt}\%}$

Water (H_2O): $\frac{2(1.008)}{2(1.008)+16.00} = \boxed{11.19\ \text{wt}\%}$

Glucose ($C_6H_{12}O_6$): $\frac{12(1.008)}{6(12.01)+12(1.008)+6(16.00)} \times 100 = \boxed{6.71\ \text{wt}\%}$

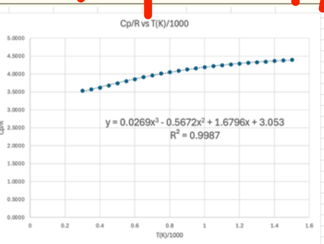
The claims regarding hydrogen storage potential of nanotubes where 1%-8% by weight was hydrogen appear to be valid because these are fairly close to the wt% hydrogen in common liquids/solids. But, the claim that hydrogen exceeded 50 wt% is unrealistic. Having this information is important so that claims like this can be validated.

P3: 15/15

Question 4: P4: 29/25

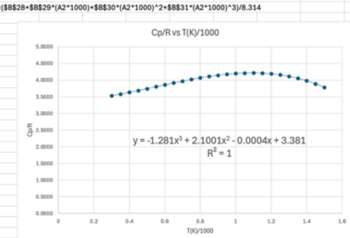
Part (a):

T(K)/1000	Cp/R (0.01 atm)
0.3	3.5345
0.35	3.5717
0.4	3.6212
0.45	3.6787
0.5	3.7396
0.55	3.8008
0.6	3.8599
0.65	3.9155
0.7	3.9672
0.75	4.0145
0.8	4.0577
0.85	4.0970
0.9	4.1327
0.95	4.1652
1	4.1948
1.05	4.2219
1.1	4.2469
1.15	4.2698
1.2	4.2912
1.25	4.3112
1.3	4.3300
1.35	4.3479
1.4	4.3651
1.45	4.3815
1.5	4.3975
A =	3.053
B=	1.6796
C=	-0.5672
D=	0.0269



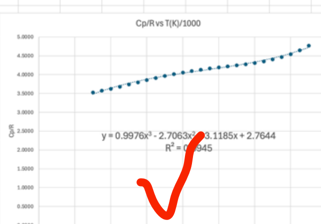
(b) i) Appendix B.17 of Murphy:

T(K)/1000	Cp/R (0.01 atm)	
0.3	3.5353	=($B$28+$B$29*(A2*1000)+$B$30*(A2*1000)^2+$B$31*(A2*1000)^3)/8.314
0.35	3.5832	
0.4	3.6349	
0.45	3.6894	
0.5	3.7457	
0.55	3.8030	
0.6	3.8601	
0.65	3.9162	
0.7	3.9704	
0.75	4.0216	
0.8	4.0689	
0.85	4.1113	
0.9	4.1479	
0.95	4.1777	
1	4.1997	
1.05	4.2130	
1.1	4.2167	
1.15	4.2097	
1.2	4.1911	
1.25	4.1600	
1.3	4.1153	
1.35	4.0562	
1.4	3.9816	
1.45	3.8906	
1.5	3.7823	
A=	28.11	
B=	-0.00000368	
C=	0.00001746	
D=	-1.065E-08	



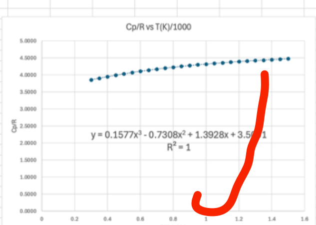
ii) Table A-21 of Moran, Shapiro, etc.

T(K)/1000	Cp/R (0.01 atm)	
0.3	3.5324	=$B$28+(A2*1000)*$B$29+$B$30*(1000*A2)^2+$B$31*(A2*1000)^3+$B$32*(A2*1000)^4
0.35	3.5753	
0.4	3.6259	
0.45	3.6816	
0.5	3.7400	
0.55	3.7992	
0.6	3.8574	
0.65	3.9133	
0.7	3.9660	
0.75	4.0145	
0.8	4.0587	
0.85	4.0984	
0.9	4.1339	
0.95	4.1658	
1	4.1950	
1.05	4.2227	
1.1	4.2505	
1.15	4.2802	
1.2	4.3141	
1.25	4.3547	
1.3	4.4048	
1.35	4.4676	
1.4	4.5467	
1.45	4.6457	
1.5	4.7690	
alpha=	3.626	
beta=	-0.001878	
gamma=	7.055E-06	
delta=	-6.764E-09	
epsilon=	2.156E-12	



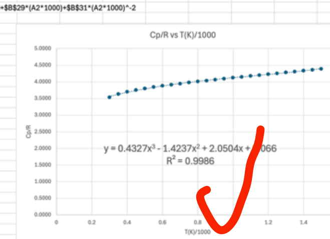
iii): Appendix B.2 of Felder + Rousseau

T(K)/1000	Cp/R (0.01 atm)	
0.3	3.8565	=($B$28+$B$29*(A2*1000)+$B$30*(1000*A2)^2+$B$31*(1000*A2)^3)/0.008314
0.35	3.9048	
0.4	3.9504	
0.45	3.9933	
0.5	4.0335	
0.55	4.0713	
0.6	4.1068	
0.65	4.1400	
0.7	4.1711	
0.75	4.2002	
0.8	4.2274	
0.85	4.2529	
0.9	4.2767	
0.95	4.2989	
1	4.3198	
1.05	4.3394	
1.1	4.3578	
1.15	4.3752	
1.2	4.3916	
1.25	4.4072	
1.3	4.4222	
1.35	4.4365	
1.4	4.4504	
1.45	4.4639	
1.5	4.4772	
a=	0.0291	
b=	0.00001158	
c=	-6.076E-09	
d=	1.311E-12	



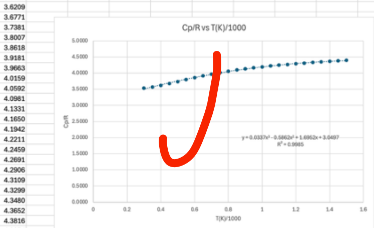
iv) Appendix C of Smith, Van Ness, Abott

T(K)/1000	Cp/R (0.01 atm)	
0.3	3.5386	=$B$28+$B$29*(A2*1000)+$B$31*(A2*1000)^-2
0.35	3.6308	
0.4	3.6995	
0.45	3.7546	
0.5	3.8012	
0.55	3.8423	
0.6	3.8795	
0.65	3.9142	
0.7	3.9469	
0.75	3.9781	
0.8	4.0083	
0.85	4.0377	
0.9	4.0664	
0.95	4.0945	
1	4.1223	
1.05	4.1497	
1.1	4.1768	
1.15	4.2037	
1.2	4.2304	
1.25	4.2570	
1.3	4.2834	
1.35	4.3096	
1.4	4.3358	
1.45	4.3619	
1.5	4.3879	
A=	3.639	
B=	0.000506	
C=	0	
D=	-22700	



v) The NIST Chemistry Webbook

T(K)/1000	Cp/R (0.01 atm)	
0.3	3.5353	=($B$28 + $B$29*A2 + $B$30*A2^2 + $B$31*A2^3 + $B$32/A2^2) / 8.314
0.35	3.5727	
0.4	3.6209	
0.45	3.6771	
0.5	3.7381	
0.55	3.8007	
0.6	3.8618	
0.65	3.9181	
0.7	3.9663	
0.75	4.0159	
0.8	4.0592	
0.85	4.0981	
0.9	4.1331	
0.95	4.1650	
1	4.1942	
1.05	4.2211	
1.1	4.2459	
1.15	4.2691	
1.2	4.2906	
1.25	4.3109	
1.3	4.3299	
1.35	4.3480	
1.4	4.3652	
1.45	4.3816	
1.5	4.3974	
	100-700 K	700-2000 K
A=	31.32234	30.03235
B=	-20.23531	8.77972
C=	57.86644	-3.988133
D=	-36.50624	0.788313
E=	-0.007374	-0.741599



Compared to the data in part (a), all coorelations show the overall trend that Cp/R increases with time, with accuracy varying. The fits from Appendix B.2, Appendix C, and NIST match closest, with Murphy showing largest deviation (mostly at higher temps.).

Question 5:

(a) A+B in 0.5 hr; A+C in 0.75 hr; B+C in 1 hr

$A+B=\frac{1}{0.5}=2$ $\quad$ $A+B+A+C=2+\frac{4}{3}$

$A+C=\frac{1}{0.75}=\frac{4}{3}$ $\quad$ $2A+\underbrace{B+C}_{1}=\frac{10}{3}$ $\quad$ $2A=\frac{10}{3}-1$ $\quad$ $A=\frac{7}{6}$ $\quad$ $\frac{7}{6}+B=2$ $\quad$ $B=\frac{5}{6}$ $\quad$ $\frac{7}{6}+C=\frac{4}{3}$ $\quad$ $C=\frac{1}{6}$

$B+C=1$

$A+B+C=\frac{7}{6}+\frac{5}{6}+\frac{1}{6}=\frac{13}{6}$ $\quad$ $\frac{1}{13/6}=\boxed{\frac{6}{13}\text{ hr } (27.7\text{ min})}$ ✓

(b) $A=\frac{1}{7/6}=\boxed{\frac{6}{7}\text{ hr } (51.4\text{ min})}$ ✓

$B=\frac{1}{5/6}=\boxed{\frac{6}{5}\text{ hr } (72\text{ min})}$ ✓

$C=\frac{1}{1/6}=\boxed{6\text{ hr } (360\text{ min})}$ ✓

Q5: 25/25